Case Study 28: Repeated Transaction Detection

1. Problem Statement

Analyze banking transaction logs to detect suspicious duplicate transactions occurring for the same account with identical amounts within a defined short time window (e.g., 60 seconds), flagging potential double-billing or fraud.

2. Objectives

- Parse and order transaction records chronologically.
- Evaluate temporal differences between consecutive transactions on identical accounts.
- Identify and flag identical-amount repeated transactions falling within the risk window.

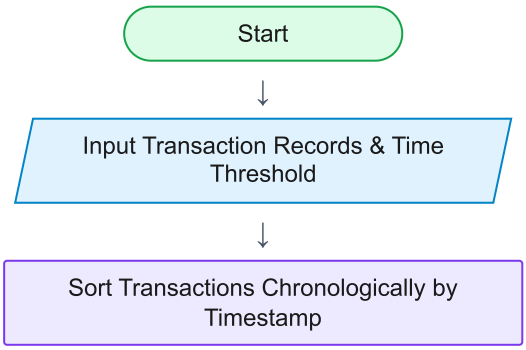
3. Assumptions

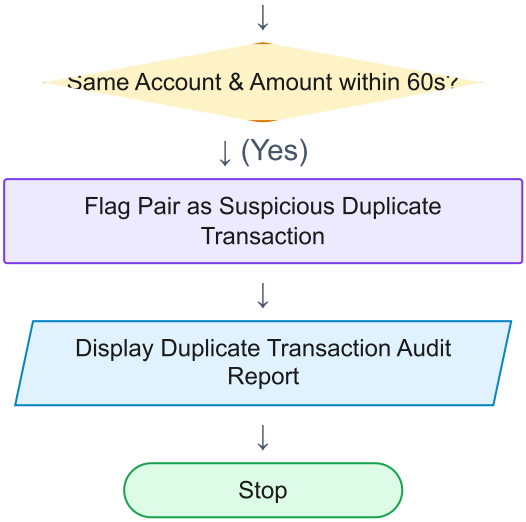
- Time threshold for duplicate flag is set to **60 seconds**.
- Transactions possess unique IDs, account identifiers, amounts, and ISO timestamps.
- A repeat transaction is defined by matching account numbers and identical dollar amounts within the threshold.

4. Algorithm

1. Start.
2. Read transaction logs and sort by timestamp in ascending order.
3. Set `time_threshold = 60` seconds.
4. For each transaction `i` from 0 to `N - 1`:
    - For each subsequent transaction `j` from `i + 1` to `N - 1`:
        - Calculate time difference: `diff = timestamp[j] - timestamp[i]`.
        - If `diff > time_threshold`, break inner loop (out of window).
        - If `account[i] == account[j]` and `amount[i] == amount[j]`:
            - Flag pair `(i, j)` as potential duplicate transaction.
5. Display flagged duplicate transactions summary report.
6. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT transactions_list threshold_seconds = 60 SORT transactions_list BY
timestamp ASCENDING flagged_duplicates = [] FOR i FROM 0 TO
LENGTH(transactions_list) - 1 DO FOR j FROM i + 1 TO LENGTH(transactions_list) - 1
DO t1 = transactions_list[i] t2 = transactions_list[j] time_diff =
SECONDS_BETWEEN(t1.timestamp, t2.timestamp) IF time_diff > threshold_seconds THEN
BREAK // Past the time window limit ENDIF IF t1.account == t2.account AND t1.amount
== t2.amount THEN APPEND (t1, t2, time_diff) TO flagged_duplicates ENDIF ENDFOR
ENDFOR PRINT flagged_duplicates STOP
```

7. Test Case and Expected Output

Evaluation Results for Account Logs:

```
Flagged Duplicate Transactions: - Flagged Tx 101 and Tx 102 for Account ACC_1
($5000) within 45.0s
```

8. Conclusion

The Repeated Transaction Detection engine accurately identifies rapid duplicate charges across transaction logs, enabling real-time alerting to prevent accidental double-billing and mitigate fraudulent activity.